

Malhar Gudekar

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EDUCATION

University of Illinois Urbana Champaign

August 2024 - May 2026

Master of Science, Information Management

GPA: 3.8/4

• Coursework: Web Programming, Machine Learning & Cloud, Information Management, Information Consulting, Methods of Data Science

University of Mumbai

June 2019 - May 2023

Bachelor of Engineering, Computer Science

GPA: 9.25/10

• Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Object-Oriented Programming, Machine Learning, NLP

SKILLS

- **Languages & DSA:** Python, SQL, C++ (basic), Data Structures & Algorithms (Arrays, Trees, Graphs, DP), LeetCode
- **Machine Learning & AI:** PyTorch, Scikit-learn, Neural Networks, NLP, Supervised & Unsupervised Learning, Feature Engineering, Generative AI, Model Optimization, Inference
- **Systems & Tools:** FastAPI, Distributed Systems, ETL Pipelines, Kafka, PySpark, PostgreSQL, Neo4j, AWS, Docker, Git

WORK EXPERIENCE

Business Intelligence Group

August 2025 - December 2025

Technical Consultant

Champaign, USA

- Built and deployed scalable, microservices-based **RAG and agentic AI system** using **FastAPI**, **PostgreSQL**, and vector embeddings for high-performance production applications
- Designed efficient, fault-tolerant **ETL, async pipelines** to process, index large document datasets for optimized low-latency retrieval systems
- Improved system performance by **64%** through optimized indexing strategies and pipeline-level tuning in distributed production environments
- Developed backend services enabling real-time, high-performance **AI model serving** of unstructured data across scalable distributed systems

University of Illinois Urbana Champaign

May 2025 - Present

Research Assistant - iSchool

Champaign, USA

- Owned **end-to-end design, development, and integration** of cross-platform **AI backend system** ingesting wearable data for 100+ users, enabling scalable data pipelines and analytics
- Designed and optimized **PostgreSQL schema, indexing strategies, and distributed query execution plans** for high-volume datasets, improving query performance by 38% in production workloads
- Built robust, production-ready **Python generative AI pipeline** with retrieval and context summarization, improving response accuracy by **43%**
- Developed **asynchronous backend workflows** for real-time, fault-tolerant processing of longitudinal health data with focus on scalability and system reliability

Research Assistant - CHI AI in Infodemic Management

January 2025 - May 2025

- Designed and deployed scalable NLP pipelines in Python, implementing supervised classification models for data preprocessing, training, and validation, reducing processing latency by 41%
- Built distributed data ingestion pipelines using **Kafka** and **Neo4j** Streams for real-time, high-throughput processing across scalable systems
- Designed and developed **production-grade REST APIs using FastAPI** to serve AI models, implementing validation, monitoring, and automated deployment workflows for reliable integration
- Enabled large-scale data processing across social media streams using **distributed systems and large-scale processing** principles for optimized performance and scalability

Swift Mobil Software Solutions Provider

July 2023 - October 2023

Data Analyst

Mumbai, IN

- Built interactive **Power BI** dashboards for **logistics** and **mobility KPIs**, reducing reporting time by **40%** and improving operational visibility
- Processed and optimized large-scale datasets using Python, NumPy, and PySpark to identify failure patterns and bottlenecks, creating visual reports to communicate insights
- Developed **SQL** and **graph queries** to map dependencies, analyze operations, detect data issues, accelerating root-cause analysis by **26%**

Pscope Technologies Pvt. Ltd.

January 2023 - June 2023

Data Analyst

Mumbai, IN

- Developed and maintained **20+ Power BI** dashboards across Sales, Finance, and Operations for enterprise clients, leveraging **SQL** and **DAX**, enabling KPI tracking and insights
- Automated recurring **Excel**-based workflows using **VBA**, implementing validation and scheduling logic, reducing report preparation time by **42%** and improving consistency

Stu/Dio at Illinois

August 2025 - Present

Project Manager

Champaign, USA

- Led cross-functional team of **6** engineers and designers to deliver **AI-driven product features** for DeepCover game, leveraging feedback and performance insights for prioritization
- Tracked **sprint** progress, delivery metrics, and **QA issues** through **JIRA**-based workflows, managing backlog and resolution for timely delivery

PROJECTS

Scalable ML System for Customer Risk Prediction

March 2025

UIUC Hackathon

- Built an end-to-end supervised ML system using Python, Pandas, NumPy, Scikit-learn, and LightGBM to train, validate, and evaluate credit risk models on large-scale time-series datasets
- Designed scalable data preprocessing and feature engineering pipelines capturing behavioral patterns, applying model validation techniques to reduce overfitting and improve generalization